



VM005 I Improved Management in Rice Production Systems, v1.0

Overview

March 11, 2025

Photo by Lisa Murray. Bale Mountains Eco-Region REDD+ Project,
Ethiopia (Verra Project 1340).

Housekeeping

- This session will last approximately 60 min.
- Q&A session at the end of the webinar (~ 10 min).
- Please put questions in the "Q&A" option in Zoom panel.
- The webinar will be recorded and available on Verra's website and YouTube channel.
- Speakers:
 - Carolina Cardoso Lisboa (Verra)
 - Sami Osman (ATOA Carbon)
 - Anna Mortimer (ATOA Carbon)

AGENDA

- INTRODUCTION
- MOTIVATION
- METHODOLOGY CONTENTS AND METHODS
- FINAL REMARKS
- Q&A

INTRODUCTION AND MOTIVATION

A photograph of a man in a forest, carrying a large log on his shoulder. He is wearing a patterned headband and a light-colored jacket. The forest is dense with green foliage and trees. The image has a blue tint. In the background, other people are visible walking along a path.

INTRODUCTION

- Verra has released [Verified Carbon Standard \(VCS\) Methodology VM005 I Improved Management in Rice Production Systems, v1.0.](#)
- VM005 I applies to projects implementing improved water and crop management practices in flooded rice systems to reduce greenhouse gas emissions.
- Verra-led methodology in consultation with [ATOA Carbon](#)
- Expert Assessment by three external experts (Expert Assessment Report(s) available on VM005 I webpage)
- A comprehensive and robust new methodology that applies the latest scientific principles to ensure the robust quantification of greenhouse emission reductions
- Wider adoption will further strengthen the credibility and integrity of rice projects and emission reduction estimates

MOTIVATION

The development of *VM0051* was motivated by the following:

- Rice is the primary food source for more than half of the global population and is cultivated over 168 million hectares of the earth's surface
- Rice fields are one of the largest sources of methane emissions in agriculture
- Verra [inactivation](#) of the UNFCCC CDM rice methodology [AMS-III.AU.: Methane emission reduction by adjusted water management practice in rice cultivation](#)
- Addressing [concerns](#) with AMS-III.AU and projects (*please visit Verra website for more information*)
- Alignment with the latest scientific understanding and recent developments in rice GHG accounting
- Future alignment with ICVCM for securing CCP labeling
- Scale the implementation of climate-friendly management practices while ensuring global food security and social-economic co-benefits

KEY FEATURES

- Safeguards to avoid soil organic carbon (SOC) losses due to the implementation of project activities
- A requirement to monitor and quantify N₂O emissions as well as CO₂ emissions from fossil fuels and energy consumption
- A dynamic baseline setting approach that accounts for potential impacts due to actual weather conditions
- Expanded guidance for project area stratification and quantification of emission reductions
- Flexible quantification approaches, including the use of biogeochemical models for emission reduction estimates
- Best practice guidance for the use of digital monitoring, reporting, and verification (DMRV), including remote sensing and machine learning/artificial intelligence, to streamline validation and verification of project data

A photograph of a man standing in a dense forest, holding a long wooden staff. He is wearing a patterned headband and a light-colored jacket. The background is filled with lush green foliage and trees. The image has a blue tint. The text "METHODOLOGY CONTENT & METHODS" is overlaid in white, bold, sans-serif font.

METHODOLOGY CONTENT & METHODS

SOURCES

- Clean Development Mechanism (CDM) AMS-III.AU Methane Emission Reduction by Adjusted Water Management Practice in Rice Cultivation, v04.0
- VM0042 Improved Agricultural Land Management, v2.1
 - Projects that seek credits for SOC stock increases (removals), employ practices that may result in material declines in SOC, or implement changes in management practices off-season changes must use VM0042

DEFINITIONS

- **AWD**

A system of cultivating irrigated lowland rice using controlled and intermittent irrigation cycles according to a periodic drainage and re-flooding schedule (i.e., alternating flooded and non-flooded field conditions by multiple (more than one) drainage events during the cultivation period) in addition to end-of-season drainage.

- **Controlled irrigation**

A type of water management where the project has irrigation systems in place that allow for controlled management of the water regime (i.e., the project farmer has control of the water flow rate, flooding duration, etc.)

- **Cultivation pattern**

The set of activities and machinery operations used to cultivate rice in the project area. This requires describing quantitative and qualitative details related to the implementation of the project activities and management practices (e.g., time, duration, rate, and frequency of wet (flooded) and dry (non-flooded) periods, machinery operations, fertilization practices, and weed and pest control)

APPLICABILITY

Project activities must:

- Identify project activity category:
 - “*Main project activities*”: improved irrigation management activities – focused on CH₄ reductions from methanogenesis
 - “*Optional project activities*”: methanotrophs, nitrogen management, biochar etc.
- Controlled irrigation / drainage
- Maintain same number of rice cultivation seasons from baseline to project
- [VCS Standard v4.7](#), requirements:
 - No clearing native ecosystems within the last 10 years

APPLICABILITY

Not applicable to:

- Management practices that decrease C-inputs to soil (*safeguards to avoid SOC losses*)
- Include upland / rainfed or deep-water and non-irrigated lowland rice
- Projects changing offseason management practices (except straw burning / irrigation)

PROJECT BOUNDARY

Source	GHG	Included?	Justification/Explanation
Soil methanogenesis	CH ₄	Yes	Changes in levels of anoxic conditions in soils lead to significant changes in soil methanogenesis. This is the dominant GHG source under this methodology.
Fossil fuels	CO ₂	S*	Where there are changes in the use of equipment such as tractors, seeders, harvesters, or irrigation pumps, these changes may result in material increases or decreases in CO ₂ emissions associated with the combustion of fossil fuels.
Liming	CO ₂	S*	Application of limestone or dolomite for soil amelioration may represent a significant source of CO ₂ .
Use of nitrogen fertilizers and nitrogen derived from crop residue (i.e., rice straw) incorporated into soils	N ₂ O	Yes	N ₂ O attributed to changes in irrigation must be included in all projects.
		S*	Where nitrogen fertilization rate and/or nitrogen returned to soils from crop residues is greater in the monitoring period than the baseline scenario, such N ₂ O emissions must be included in the project boundary.
Biomass burning	CH ₄	S*	Biomass burning releases CH ₄ .
	N ₂ O	S*	

S* – Must be included where the project activity significantly increases total GHG emissions by more than 5% compared to the baseline scenario (i.e., emissions are not de minimis) and may be included where the project activity reduces total GHG emissions by more than 5% compared to the baseline scenario.

BASELINE SCENARIO

ALM Practices	Qualitative	Quantitative
Water regime (pre- and on-season)	• Irrigation (Y/N) • Flooding on-season (cont. / single / multiple drainage) • Flooding pre-season (flooded / short / long drainage)	• Irrigation (rate / volume) • Flooding duration (days) • Water level (below surface)
Crop planting and harvesting	• Crop types • Rice variety • Rice seeding practice (DSR / transplanting)	• Approx seeding date • Approx. harvest / termination date • Crop yield
Tillage and/or residue management	• Tillage (Y/N) • Crop residue removal (Y/N) • Crop residue incorporation (Y/N) • Burning crop residues (Y/N) • Usage of non-burned crop residues	• Tillage depth • Tillage frequency • Tillage area • Crop residue rate left in situ • Crop residue incorporation timing • Burned area
Nitrogen fertilizer application	• Organic amendment (Y/N) • Org amendment type(s) • Synthetic N fertilizer (Y/N) • Synthetic fertilizer type(s)	• Organic amendment type & application rate • Synthetic fertilizer type & application rate
Other amendments	• Biochar (Y/N) • Methanotrophs (Y/N)	• Biochar application rate • Methanotrophs application rate
Liming	• Application of calcitic limestone or dolomite	• Calcitic limestone or dolomite rate & dates of application
Machinery and equipment operations	• Fossil fuel usage (Y/N) • Fossil fuel type(s)	• Fossil fuel type / rate

ADDITIONALITY

Project Method – aligned with VCS Additionality Tool - [VT0008](#)

- Regulatory Surplus
- Barrier Analysis and/or Investment Analysis
- Common Practice Analysis
 - 20% penetration rate threshold
 - Alternative evidentiary sourcing
 - Gov data - Peer reviewed literature – independent research data – attestations (e.g., from experts in the field) – grower surveys – remote sensing datasets

QUANTIFICATION OF REDUCTIONS

Three Quantification Approaches

- QAI – Modeling
 - Must follow VMD0053 / VM0042 guidance – ensure calibration/validation for rice and undertake initial SOC measurements within the project area
 - Any SSRs that cannot be modeled must be covered by defaults (i.e. CO₂ liming, fossil fuels)

QUANTIFICATION OF REDUCTIONS

- QA2 – Direct Measurement of CH₄
 - Chamber measurements and chromatography analysis
 - Requires field measurements at baseline control sites
 - Compulsory for projects implementing: Methanotrophs, biochar, Low emission rice varieties, AWD drainage at a depth < 10cm
 - All other SSRs must be covered by modeling or defaults
- QA3 – Default Emission Factors
 - Tier I for projects with a capacity limit of 60 000 tCO₂e per year

QUANTIFICATION OF REDUCTIONS

GHG	Source	Quantification Approach 1: Modeling	Quantification Approach 2: Direct Measurement	Quantification Approach 3: Default Emission Factors
CH ₄	Soil methanogenesis	X	X	X
CH ₄ , N ₂ O	Biomass burning			X
N ₂ O	Use of nitrogen fertilizers and nitrogen from rice straw	X		X
CO ₂	Liming			X
CO ₂	Fossil fuels			X
CO ₂	SOC	X		

QUANTIFICATION OF REDUCTIONS - Uncertainty

- QA 1: Modeling
 - Apply uncertainty requirements in VMD0053
 - Focus on Model prediction error, Sampling error & Measurement error
 - (i) Analytical calculation of error propagation;
 - (ii) Monte Carlo simulation
- QA 2: Direct Measurement
 - Focus on Sampling Error and Measurement Error

QUANTIFICATION OF REDUCTIONS - Uncertainty

- QA 3: Default Emission Factors
 - Tier I EF, projects with a capacity limit of 60 000 tCO₂e per year
 - Default 15% uncertainty deduction
 - Project with a capacity limit > 60 000 tCO₂e per year:
 - Derive uncertainty estimates from source literature or calculate using:
 - (i) Analytical calculation of error propagation;
 - (ii) Monte Carlo simulation

QUANTIFICATION OF REDUCTIONS

Leakage

Accounted for in 3 separate ways:

- New application of organic amendments from outside the project area
- Declines in rice yield; and/or
- Diversion of biomass residues used for bioenergy applications pre-project

Guidance replicates VM0042 requirements for each

Any leakage source can be ignored if de minimis (less than 5% of total reductions of the project)

MONITORING

- Guidance on sourcing data
 - Box 1. Details on the selection of preferred sources
- Data requirements for baseline
 - Table 3. Minimum specifications for the identification of agricultural land management (ALM) practices in rice production systems in the baseline scenario - see slide 14

MONITORING

- Data requirements for additionality
 - Agricultural census or other government (e.g., survey) data
 - Peer-reviewed scientific literature
 - Independent research data
 - Attestation statement from a qualified independent local expert (e.g., accredited experts in the field)
 - Grower surveys conducted within the project region
 - Reports or assessments compiled by industry associations
 - Data compiled using remote sensing datasets

Appendixes

- Appendix 1 and 2 - QA2 Direct Measurement
 - Suitable methane analysis equipment and techniques
 - Minimum experience requirements for staff performing key roles
 - Minimum requirements for stratification/sample plan design
 - QA/QC for flux measurements
 - Accreditation for laboratories

Appendixes

- Appendix 4. Digital Monitoring, Reporting & Verification - DMRV
 - DMRV is encouraged – in particular remote sensing – to efficiently enable third-party validation and verification project data
 - Appendix 4 provides extensive guidance re QA/QC best practices for using DMRV for this methodology

FINAL REMARKS



FINAL REMARKS

VM005I

- With this new methodology, project proponents can credibly achieve emission reductions and generate high-quality [Verified Carbon Units \(VCUs\)](#)
- Verra already submitted VM005I for approval under [ICVCM's CCP assessment framework](#) and [ICAO's CORSIA Eligible Emission Units](#)
- Verra encourages proponents to use the digital version of the methodology (*available soon*)
- Verra will explore adaptation of VM005I in to the new Scope 3 Standard (S3S) Program ([public consultation](#))

A man in a green jacket and a purple headwrap is walking through a dense forest, carrying a large piece of wood on his shoulder. In the background, other people are visible walking along a path. The scene is set in a lush, green forest with large trees and dense foliage.

Q&A

For general questions pertaining to the Verra and the VCS Program: info@verra.org

For project related questions: secretariat@verra.org include project ID



THANK YOU

CAROLINA CARDOSO LISBOA
MANAGER
PROGRAM DEVELOPMENT AND INNOVATION DEPARTMENT
VERRA.ORG
INFO@VERRA.ORG

SAMI OSMAN
CHIEF EXECUTIVE OFFICER
ATOACARBON
ATOACarbon.COM
SAMI@ATOACARBON.COM

